

# Genome-wide analysis of the stress associated protein (SAP) gene family containing A20/AN1 zinc-finger(s) in rice and their phylogenetic relationship with *Arabidopsis*

Shubha Vij · Akhilesh K. Tyagi

Received: 18 July 2006 / Accepted: 31 August 2006 / Published online: 11 October 2006  
© Springer-Verlag 2006

**Abstract** Proteins with the A20/AN1 zinc-finger domain are present in all eukaryotes and are well characterized in animals, but little is known about their function in plants. Earlier, we have identified an A20/AN1 zinc-finger containing stress associated protein 1 gene (*SAP1*) in rice and validated its function in abiotic stress tolerance. In this study, genome-wide survey of genes encoding proteins possessing A20/AN1 zinc-finger, named *SAP* gene family, has been carried out in rice and *Arabidopsis*. The genomic distribution and gene architecture as well as domain structure and phylogenetic relationship of encoded proteins numbering 18 and 14 in rice and *Arabidopsis*, respectively, have been studied. Expression analysis of the rice *SAP* family was done to investigate their response under abiotic stress conditions. All the genes were inducible by one or the other abiotic stresses indicating that the *OsSAP* gene family is an important component of stress response in rice. Manipulation of their expression and identification of their superior alleles should help confer stress tolerance in target crops.

**Keywords** A20 · Abiotic stress · AN1 · *Arabidopsis* · Expression analysis · Rice · Stress associated protein · Zinc-finger

## Introduction

The A20 zinc-finger domain was first identified in a TNF $\alpha$ -inducible protein in human endothelial cells (Dixit et al. 1990; Oipari et al. 1990). This domain is present in the C-terminus part of the protein and is characterized by multiple Cys<sub>2</sub>/Cys<sub>2</sub> finger motifs (Oipari et al. 1990). This protein plays an important role in regulating the immune response by inhibition of NF $\kappa$ B activity, which is an important transcriptional regulator involved in innate and adaptive immunity, inflammation, development, cell proliferation and survival. Moreover, overexpression of this protein could inhibit TNF-mediated cell death in a number of cell types (Heyninck and Beyaert 2005). The AN1 domain was first identified as a putative zinc-finger domain in the proteins coded by the *Xenopus laevis* animal hemisphere 1 (AN1) maternal RNA (Linnen et al. 1993). The AN1 zinc-finger domain is usually found associated with the A20 zinc-finger and such type of proteins are present in all eukaryotes (Evans et al. 2004). In addition to the human A20 zinc-finger protein, similar proteins such as ZNF216 and AWP1 from humans (Scott et al. 1998; Duan et al. 2000) and ZNF216 from mouse (Hishiya et al. 2006) have been characterized. The human ZNF216 has the A20 zinc-finger at the N-terminus and the AN1 zinc-finger at the C-terminus. Similar to the human A20 protein, ZNF216 also negatively regulates the NF $\kappa$ B activation pathway and also interacts with components of the immune response like

Communicated by K. Shirasu.

**Electronic supplementary material** Supplementary material is available in the online version of this article at <http://dx.doi.org/10.1007/s00438-006-0165-1> and is accessible for authorized users.

S. Vij · A. K. Tyagi (✉)  
Interdisciplinary Centre for Plant Genomics and  
Department of Plant Molecular Biology,  
University of Delhi South Campus,  
Benito Juarez Road,  
New Delhi 110021, India  
e-mail: akhilesh@genomeindia.org

RIP, IKK $\gamma$  and TRAF6. Interaction with IKK $\gamma$  and RIP is mediated by the A20 zinc-finger domain and the AN1 zinc-finger domain mediates its interaction with TRAF6, showing that both the zinc-finger domains are involved in regulating the immune response (Huang et al. 2004). The *indica* rice OsiSAP1 was identified as the first plant protein having zinc-fingers (A20 and AN1) present at the N- and C-terminus, respectively, and its association with multiple stresses was unraveled (accession no. AF140722.1, 1999; AY137590.1, 2002) (Mukhopadhyay et al. 2004). It showed homology to a previously reported pathogenesis-related protein from bean (Sharma et al. 1992) and its homolog from *japonica* rice was used to establish protein level network (Cooper et al. 2003). *OsiSAP1* is induced in response to multiple environmental stresses like cold, salt, drought, submergence, wounding and heavy metals. In addition, overexpression of *OsiSAP1* conferred abiotic stress tolerance in transgenic tobacco (Mukhopadhyay et al. 2004). Although the role of some of the A20/AN1 zinc-finger proteins in regulating the immune response has been well worked out in animal systems, more such information is required about the function of these proteins in plants. In fact, to the best of our knowledge, only *indica* rice *OsiSAP1* has been characterized for its function in abiotic stress response.

In order to understand the role of the *SAP* gene family in plants, a genome-wide survey has been done using the finished level rice and *Arabidopsis* genome sequence (The Arabidopsis Genome Initiative 2000; The International Rice Genome Sequencing Project 2005; Vij et al. 2006) for identification of *SAP1* paralogs/orthologs in rice and *Arabidopsis*. The phylogenetic relationship and domain organization of the identified rice (18) and *Arabidopsis* (14) proteins was also done. In addition, the role of *SAP* gene family in abiotic stress conditions was established by expression profiling under abiotic stress conditions.

## Materials and methods

### In silico analysis

Putative A20/AN1 domain containing proteins were extracted from the NCBI protein database using ‘A20 zinc-finger’ and ‘AN1 zinc-finger’ as keywords. In total, 143 A20 and 161 AN1 zinc-finger domain containing proteins were obtained. Comparison of the two sets of sequence revealed that the 143 sequences retrieved giving ‘A20 zinc-finger’ as keyword were also represented in the ‘AN1 zinc-finger’ sequence subset, hence the 161 sequences retrieved using AN1 zinc-finger as

key word were used to generate a HMM model (<http://www.hmmerr.wustl.edu>; Eddy 1998). The putative *SAP* gene family members, encoding A20/AN1 domain, in the rice genome were identified by using the HMM model for searching against version 4 of the TIGR rice pseudomolecules (Chromosomes 1–12) (<http://www.tigr.org/tdb/e2k1/osa1/pseudomolecules/info.shtml>) and against the KOME cDNA database (<http://www.cdna01.dna.affrc.go.jp/cDNA/>). The sequences retrieved from this search were aligned using ClustalX multiple alignment program (Thompson et al. 1997) to identify unique sequences. In addition, the TIGR rice genome browser ([http://www.tigr.org/tigr-scripts/osa1\\_web/gbrowse/rice/](http://www.tigr.org/tigr-scripts/osa1_web/gbrowse/rice/)) was used for verifying the alignment of each TIGR gene loci with the KOME rice full-length cDNA (if present). The *SAP* gene family members present in the *Arabidopsis* genome were identified by using the HMM model for search against the TAIR database (<http://www.Arabidopsis.org/>). Unique A20/AN1 sequences were identified by using ClustalX multiple alignment program (Thompson et al. 1997). The nucleotide length, number of introns, predicted amino acid length and chromosomal location of each rice and *Arabidopsis* gene were obtained from TIGR (<http://www.tigr.org/>). These members of the *SAP* gene family were identified with numbers specific to a plant species and initials of genera and species (e.g. *Os* or *At*) or subspecies (e.g. *i* for *indica* and *j* for *japonica*) were added to delineate their origin, wherever necessary.

For identifying rice genes present in duplicated regions, each gene was searched in the regions of segmental duplication (distance = 100 kb) identified in the TIGR rice pseudomolecules ([http://www.tigr.org/tdb/e2k1/osa1/segmental\\_dup/index.shtml](http://www.tigr.org/tdb/e2k1/osa1/segmental_dup/index.shtml)). *Arabidopsis* *SAP* genes present in duplicated regions were identified by searching the ‘Paralogons in *Arabidopsis thaliana*’ website (<http://www.wolfe.gen.tcd.ie/athal/dup>) using the TAIR accession numbers. Only those *Arabidopsis* genes were considered to be duplicated which were falling in duplication blocks with size greater than or equal to 5.

The presence of A20 and/or AN1 zinc-finger domain, their length and organization in each sequence was characterized by domain search in the SMART (<http://www.smart.embl-heidelberg.de/>) and InterPro database (<http://www.ebi.ac.uk/interpro/>).

The 1 kb region upstream of each gene was searched for abiotic stress associated elements in the PLACE database (Higo et al. 1999).

### Plant materials and growth conditions

For quantitative real-time PCR analysis, rice seeds (*Oryza sativa* subsp. *indica* var Pusa Basmati 1) were

taken in a plastic beaker and sterilized with 0.1% mercuric chloride and few drops of detergent for 10 min. After sterilization, seeds were washed at least five times with tap water and soaked in dark overnight. The following day, seeds were evenly spread on a 5–6 cm thick cotton bed soaked with water. The tray was covered with a humidity chamber (made by wrapping cling film over an aluminum frame). The seeds were allowed to germinate in the culture room maintained at  $28 \pm 1^\circ\text{C}$  with a 16 h/8 h light/dark cycle. The source of light were fluorescent tubes (Philips Champion 40W/54, Philips India Limited, India) providing light at a fluence of  $50\text{--}100 \mu\text{mol m}^{-2}\text{s}^{-1}$ . For each kind of stress treatment, 7-day-old rice seedlings were subjected to cold ( $4^\circ\text{C}$ ), salt (200 mM NaCl) and dehydration (dried on tissue paper) for 6 h.

#### Quantitative real-time PCR analysis

RNA was isolated using RNeasy Mini Kit (Qiagen, Germany) as per manufacturer's instructions. The quality and quantity of each RNA sample was checked by agarose gel electrophoresis. First strand cDNA was synthesized from 1  $\mu\text{g}$  total RNA using high capacity cDNA archive kit (Applied Biosystems, USA) as per manufacturer's instructions. Details of sequences of primers used for real time PCR are given in Supplementary Table 1. In order to confirm that the expression pattern obtained was gene specific, the primers were designed outside of the conserved region. Additionally, each primer pair was also verified for uniqueness by BLAST search against the NCBI nucleotide database. Uniqueness of the expression data was also confirmed by visualizing the dissociation curve after the real-time PCR reaction. For setting-up the real time PCR reaction, diluted cDNA sample was used as a template for amplification using 200 nM each of forward and reverse primer and PCR SYBR green reaction mix (Applied Biosystems). Each reaction mix was loaded in triplicate in a 96-well plate for each biological replicate. The reaction cycle was as follows:  $50^\circ\text{C}$  (2 min)/ $95^\circ\text{C}$  (10 min)/ $95^\circ\text{C}$  (15 s)/ $60^\circ\text{C}$  (1 min) X40 cycles]. In each case, *ACTIN* was used as the internal control. At the end of the reaction, threshold CT value was obtained for each sample. The CT value of *ACTIN* was subtracted from CT value of sample to get  $\Delta\text{CT}$  value. In each case, the  $\Delta\text{CT}$  value of unstressed samples was taken as the calibrator value. The relative mRNA value in each case was calculated using the formula: Relative mRNA level =  $2^{-\Delta\text{CT}_{\text{sample}} - \Delta\text{CT}_{\text{calibrator}}}$ . At least two biological replicates were taken in each case.

## Results

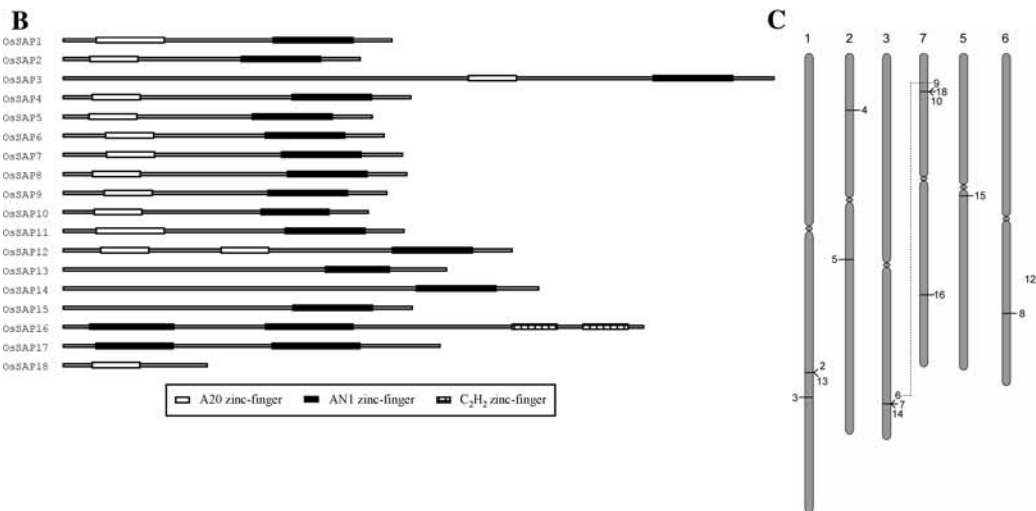
### The *SAP* gene family in rice

The HMM profile generated using the 161 A20/AN1 sequences retrieved from the NCBI protein database was used for homology search against the KOME database and version 4 of TIGR pseudomolecules. This search identified 18 *SAP* gene family members in the rice genome (Table 1). Out of the 18 OsSAP proteins (named after OsSAP1; Mukhopadhyay et al. 2004), 12 (OsSAP1–12) had both the A20 (present at the N-terminus) and the AN1 (present at the C-terminus) zinc-finger domains. OsSAP12 was an exception with two A20 zinc-finger domains present at the N-terminus instead of one. OsSAP13–15 as well as OsSAP16–18 were atypical *SAP* family members. OsSAP13–15 had only one AN1 zinc-finger domain while OsSAP16–17 had 2 AN1 zinc-finger domains. OsSAP18 was the only family member where the AN1 zinc-finger domain was absent and only the A20 zinc-finger domain was present (Fig. 1a, b). Study of the intron–exon organization of the *SAP* family showed that 11 of the genes (more than 60%) did not have any introns, six had a single intron and only 1 family member (*OsSAP8*) had 2 introns (Table 1).

Members of the rice *SAP* gene family were mapped to the TIGR rice pseudomolecules (version 4; chromosomes 1–12) using the coordinates given for each locus in TIGR (<http://www.tigr.org/tdb/e2k1/osa1/pseudomolecules/info.shtml>). The members of *SAP* gene family were found to be present on chromosomes 1, 2, 3, 5, 6, 7, 8 and 9 of the rice genome. The majority of the genes were present on chromosome 7 (four), while chromosome 5 and 6 had only one gene each. Regions of the rice genome with segmental duplications available at TIGR ([http://www.tigr.org/tdb/e2k1/osa1/segmental\\_dup/index.shtml](http://www.tigr.org/tdb/e2k1/osa1/segmental_dup/index.shtml)) were searched in a 100 kb window for the presence of *SAP* gene family members. Two pairs of genes were identified in the duplicated segments of the rice genome. *OsSAP1* and *OsSAP11* are present on chromosomes 9 and 8, respectively and *OsSAP6* and *OsSAP9* are present on chromosomes 3 and 7, respectively (Fig. 1c).

### The *SAP* gene family in *Arabidopsis*

The HMM profile was also used to search against the TAIR database to find A20/AN1 zinc-finger encoding *SAP* gene family members present in the *Arabidopsis* genome. This search identified 14 *SAP* family members in the *Arabidopsis* genome, out of which 10 (*AtSAP1*–10) code both the A20 and the AN1 zinc-finger





**Fig. 1** The rice SAP family. **a** Alignment of rice *SAP* gene family encoded A20/AN1 zinc-finger proteins using clustal method. The conserved cysteine residues characteristic of the A20 zinc-finger are highlighted by medium grey colour and the conserved cysteine and histidine residues characteristic of the AN1 zinc-finger domain are marked in dark grey colour. Residues showing  $\geq 50\%$  conservation are marked in light grey colour. Gaps indicated by dashes have been introduced to maximize alignment. The first 100 aa of OsSAP3 and the last 64 aa of OsSAP16 have not been shown as they did not align with any of the other sequences. **b** Domain organization of the rice *SAP* gene family encoding A20/AN1 zinc-finger proteins. The size of each domain was obtained

from SMART (<http://www.smart.embl-heidelberg.de/>) and InterPro database (<http://www.ebi.ac.uk/interpro/>) and is to scale with respect to the protein size. Note that OsSAP1-12 have both the A20 and AN1 zinc-finger domains, OsSAP14-17 have only the AN1, whereas OsSAP18 has only the A20 zinc-finger domain. **c** Chromosomal localization of the rice *SAP* gene family members. The numbers indicate the gene identity given in Table 1. Genes present in duplicated segments are connected by dashed lines. Note that rice chromosomes 4, 10, 11 and 12 are not shown since A20/AN1 domain encoding genes were not present on these chromosomes

domains. AtSAP11–13 had two AN1 zinc-finger domains each while AtSAP14 had only a single AN1 zinc-finger domain. Unlike rice, there was no *Arabidopsis* SAP family member which had only the A20 zinc-finger domain (Fig. 2a, b). The majority (more than 60%) of the *SAP* gene family members had a single intron, three had no intron (*AtSAP4*, 5 and 10). *AtSAP2* had 2 introns while *AtSAP14* had 3 introns (Table 2).

The location of *Arabidopsis* A20/AN1 zinc-finger genes was mapped to the *Arabidopsis* genome (<http://www.Arabidopsis.org>). The genes were distributed on all chromosomes. The majority of the genes were present on chromosomes 3 and 4 (4 each) while chromosome 5 had only one gene. Six of the fourteen *Arabidopsis* genes were identified in duplicated blocks (Fig. 2c).

### Phylogenetic analysis

For the purpose of phylogenetic analysis, 18 rice and 14 *Arabidopsis* protein sequences were aligned using

ClustalX multiple alignment program. An unrooted phylogenetic tree was constructed for *Arabidopsis* and rice using the neighbor-joining method and viewed using Treeview. The rice and *Arabidopsis* SAP family members formed six distinct groups (Fig. 3). This included two *Arabidopsis* specific groups, group IV (*AtSAP3*, 4 and 6) and group V (*AtSAP1*, 7, 8, 9, 10) and two rice specific groups, group II (OsSAP10, OsSAP18, OsSAP12, OsSAP7 and OsSAP14) and VI (OsSAP3, 5). Of the remaining two groups containing both rice and *Arabidopsis* proteins, group I contained majority of proteins with only the AN1 zinc-finger domain (Fig. 3).

The rice *SAP* gene family members are involved in abiotic stress response

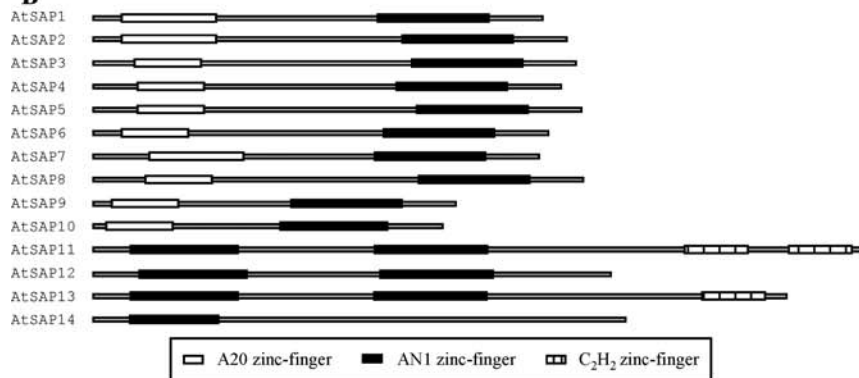
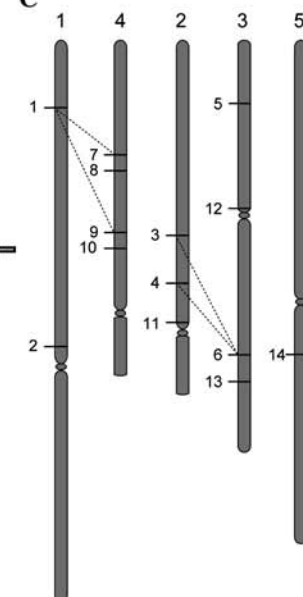
Real time PCR analysis was used to evaluate the response of the rice *SAP* gene family encoding A20/AN1 zinc-finger proteins to abiotic stress. RNA was isolated from 7-day-old rice seedlings, which were

**Table 1** Distribution of *OsSAP* gene family encoding A20/AN1 zinc-finger proteins in the rice genome

| S.No | Zinc-finger       | Gene_ID        | Locus ID_TIGR    | KOME     | Chr | Coordinates (5' to 3') | Nucleotide length (bp) | Predicted protein (aa) | No. of introns |
|------|-------------------|----------------|------------------|----------|-----|------------------------|------------------------|------------------------|----------------|
| 1    | A20-AN1           | <i>OsSAP1</i>  | LOC_Os09g31200.1 | AK106128 | 9   | 18,759,705–18,760,622  | 495                    | 165                    | 0              |
| 2    | A20-AN1           | <i>OsSAP2</i>  | LOC_Os01g52030.1 | AK111476 | 1   | 30,231,637–30,230,953  | 447                    | 149                    | 0              |
| 3    | A20-AN1           | <i>OsSAP3</i>  | LOC_Os01g56040.1 |          | 1   | 32,593,903–32,596,306  | 1068                   | 356                    | 1              |
| 4    | A20-AN1           | <i>OsSAP4</i>  | LOC_Os02g10200.1 | AK067403 | 2   | 5,347,486–5,345,611    | 522                    | 174                    | 1              |
| 5    | A20-AN1           | <i>OsSAP5</i>  | LOC_Os02g32840.1 | AK063708 | 2   | 19,482,320–19,483,065  | 465                    | 155                    | 0              |
| 6    | A20-AN1           | <i>OsSAP6</i>  | LOC_Os03g57890.1 | AK102503 | 3   | 32,915,847–32,913,214  | 483                    | 161                    | 1              |
| 7    | A20-AN1           | <i>OsSAP7</i>  | LOC_Os03g57900.1 | AK108205 | 3   | 32,921,903–32,921,106  | 510                    | 170                    | 0              |
| 8    | A20-AN1           | <i>OsSAP8</i>  | LOC_Os06g41010.1 | AK069720 | 6   | 24,490,982–24,493,135  | 516                    | 172                    | 2              |
| 9    | A20-AN1           | <i>OsSAP9</i>  | LOC_Os07g07350.1 | AK121813 | 7   | 3,676,709–3,674,265    | 486                    | 162                    | 1              |
| 10   | A20-AN1           | <i>OsSAP10</i> | LOC_Os07g07400.1 |          | 7   | 3,697,428–3,696,970    | 459                    | 153                    | 0              |
| 11   | A20-AN1           | <i>OsSAP11</i> | LOC_Os08g39450.1 | AK101969 | 8   | 24,821,094–24,822,077  | 513                    | 171                    | 0              |
| 12   | A20-A20-AN1       | <i>OsSAP12</i> | LOC_Os08g33880.1 |          | 8   | 21,072,646–21,073,320  | 675                    | 225                    | 0              |
| 13   | AN1               | <i>OsSAP13</i> | LOC_Os01g51990.1 |          | 1   | 30,207,127–30,205,282  | 576                    | 192                    | 1              |
| 14   | AN1               | <i>OsSAP14</i> | LOC_Os03g57920.1 | AK108384 | 3   | 32,927,052–32,925,978  | 714                    | 238                    | 0              |
| 15   | AN1               | <i>OsSAP15</i> | LOC_Os05g23470.1 | AK072411 | 5   | 1,3345,932–13,343,520  | 525                    | 175                    | 0              |
| 16   | AN1-AN1-C2H2-C2H2 | <i>OsSAP16</i> | LOC_Os07g38240.1 |          | 7   | 22,929,069–2,2931,587  | 873                    | 291                    | 1              |
| 17   | AN1-AN1           | <i>OsSAP17</i> | LOC_Os09g21710.1 |          | 9   | 13,154,831–13,153,949  | 567                    | 189                    | 0              |
| 18   | A20               | <i>OsSAP18</i> | LOC_Os07g07370.1 |          | 7   | 3,683,966–3,683,751    | 216                    | 72                     | 0              |

**A**

|         |                                                                                |     |
|---------|--------------------------------------------------------------------------------|-----|
| AtSAP1  | ---MGSEQNDSTSFSP---SEPKLVKCGGFFGSPSNMNLCSKCYRDIRATE---EQTASAKAAVEKSLN-----     | 61  |
| AtSAP2  | ---MDHDKTGCQS---PPEGPKLTNNCGFFGSAATMMCSKCHKDMLFQEEQGAFFASAVSGTSSSSNIKET---     | 68  |
| AtSAP3  | ---MAEHRQLQEP---RLGANNCGFFGSTATQNLCSKCFRDLQHQE---QNS-STAKHALTQSLAAVG---        | 58  |
| AtSAP4  | ---MAEHRCEPTE---GHRLENNCGFFGSSATMNLCSNCGDLCLKQ---QQQ-ASMKSTVESSL-----          | 57  |
| AtSAP5  | MAQRTEKEETEFKVLETLTTTTTTTNNCGVTANPATNMCKQCFN---ASLVSAAGVVEGSGILKR---           | 66  |
| AtSAP6  | ---MAEHRCEPTE---SNRLGVNCGFFGSSATMNLCSNCGDLCLKQ---QQQSSSIKSTVESSLVSPSSSS        | 67  |
| AtSAP7  | ---MGSENNSTSFPP---TEPKLDNCGGFFGSPSNMNLCSKCYRSIRAE---DQTAVAKAAVKNLKL---LPSCSI   | 67  |
| AtSAP8  | ---MTGE---PSLIRGCGFFSTSTQTKNLCSKCYNDLFKDE---SARYLATFNVN---                     | 46  |
| AtSAP9  | ---MGSEQNDSTSFQSQASEPKLVKCGGFFGSPSNMNLCSKCYRGICAE---AQTAVAKAAVEKSKPKSPRSLF     | 72  |
| AtSAP10 | ---MVNTEALP---EGGGLYGRVNNNLCSLYKKSVLQH---SPALRFEPETECS---                      | 50  |
| AtSAP11 | ---MGTEPFPPDLG---KHCSVDVQKIDFLPFTCDRLQVLCILHRSYMK---AS---CPKGDR                | 52  |
| AtSAP12 | ---MAGGTEAFPPDLG---EHCQDPDCKLLDLPFTCDCKLVCEHRSYKS---HN---CPKSDH                | 55  |
| AtSAP13 | ---MGTEPFPPDLG---KHCSVDVQKIDFLPFTCDRLQVLCILHRSYMK---HD---CPKGNR                | 52  |
| AtSAP14 | ---MGTEPFPPDLG---KHCSVDVQKIDFLPFTCDRLQVLCILHRSYMLNPDEDSNITWDKHNVDTLCPSPNY      | 68  |
| AtSAP1  | -PNKPKTQPQQSQEITQVGLSGSSSSSTRGGDSAAA-----PLDPPK                                | 103 |
| AtSAP2  | -FTALVDIETKSVFMTVSVPSSVQVVAEVAP-----EEAAKP                                     | 108 |
| AtSAP3  | -AAASSSVSPPPPPADSKEIVEAK--SEKRAAAEPE-----EADGPP                                | 98  |
| AtSAP4  | -SPVIAPVLENYAAELEIPTTKTEKKPIQIPT-----EQSPSP                                    | 96  |
| AtSAP5  | -SARSVNLRSS--PAKVIRPREIDAVKK--RD-----QQIVN                                     | 99  |
| AtSAP6  | SEISSPIIPPLLKNPSVKLEVP-----EKKAVISLPT-----TEQNQQ                               | 105 |
| AtSAP7  | IAPGQKHPLKIPAHLETVVTAEPSPVPAEQDEA-----EPSPRV                                   | 110 |
| AtSAP8  | ---TKAAEVTAEATVLGS-----                                                        | 63  |
| AtSAP9  | IAEPPAVVVEPKPEKAAVVVSAEPSSSAVPEAN-----EPSPRA                                   | 112 |
| AtSAP10 | ---QCCPPTNSPAVEEPPVKK-----                                                     | 68  |
| AtSAP11 | EDVTVVICPLCAGVRLNPEDPNIT--WEKHVN-TDCDPSNYEKATKKKKVPPRCKEYLTFSNTIKRDRGNVDHLKH   | 129 |
| AtSAP12 | GSRTVSICTCSIAIETTGFDEKGIKSLLEKHERSGDCDPN--KKKK-PTCEPVKRCIEILTANNLTCKYGVKPKLKH  | 131 |
| AtSAP13 | GDVTVVICPLCAGVRLNPEDPNIT--WEKHVN-TDCDPSNYEKAVKKKKVPPRCRELLTFSNTIKRDRGNVDHLKH   | 129 |
| AtSAP14 | ENDVKKKK--KQCPVS-----SCSGPKEPD--SSISNTNTVAATSAP-----                           | 107 |
| AtSAP1  | ST-----ATRLCSNKKVGVTFGFCRCGCTGCTHRYPEHSCDFDK-----                              | 146 |
| AtSAP2  | KG-----PSRCTTCKRRVGLTGFCRCGSLGCTHRYADVDGDSFNH-----                             | 151 |
| AtSAP3  | QD-----PKRCLTGRRRVGITGFCRCGTFVCGTHRYAFCGECDFDK-----                            | 141 |
| AtSAP4  | QR-----PNRCVTCKRRVGLTGFCRCGTTGCTHRYPEHVGCTEDFK-----                            | 139 |
| AtSAP5  | -----RCSGCRKKVGLTGFCRCGELGCEHRYSDRHDGSDYDK-----                                | 138 |
| AtSAP6  | QR-----PNRCVTCKRRVGLTGFCRCGTMCGVHRYPEHSCSDYDK-----                             | 148 |
| AtSAP7  | RP-----NNRCFSNKKVGVGMFGFCRCGCTGCTHRYPEKHEGSDYDK-----                           | 153 |
| AtSAP8  | -----KGGCACKKVGLLGHFRCGHLFASHRYPEHSCSDYDK-----                                 | 103 |
| AtSAP9  | R-----TNRCCLCNKKVGMFGFCRCGCTGCTHRYPETEDGSDYDK-----                             | 154 |
| AtSAP10 | -----RCGICRKKVGMFGFCRCGTMCGVHRYPEHSCSDYDK-----                                 | 105 |
| AtSAP11 | RFGPDLKCPGPRK---LPFMGLSSSTTRKEAKTTRPNKAHPSTSSSSSSSRWSNLLSSAEAGI--SRLGNDISQKLQF | 203 |
| AtSAP12 | RFPTTHVCKNKKIINTAGTSSRWN---ERFMEALSLRNQKCGRGSSVSSKSSPSVRSF-----                | 186 |
| AtSAP13 | RFGPIHSSGPKPESSFSFMGLSTNT--KEAP-----ASSSSSRWSLFAAEASI--SRLGNDISQKLQF           | 194 |
| AtSAP14 | -ASSSSSSISSASFFASAEARFKTRLNRT---LNRRLNPTRED-----                               | 149 |
| AtSAP1  | ---GVAREAIKANPVVKADKVDRI-----                                                  | 168 |
| AtSAP2  | ---AAAEAIKANPVVKAELDKI-----                                                    | 173 |
| AtSAP3  | ---RMGDKIAKANPIVKADEKLEKI-----                                                 | 163 |
| AtSAP4  | ---SAGREEIAKANPLVIAKLQKI-----                                                  | 161 |
| AtSAP5  | ---TAGREEIARENPVVKAAMVKV-----                                                  | 160 |
| AtSAP6  | ---SAGREEIAKANPLVKAALQKI-----                                                  | 170 |
| AtSAP7  | ---EVGRDAIAKANPLVKAADKQRI-----                                                 | 175 |
| AtSAP8  | ---SAAIDVLAKONPVVKGDKLFR-----                                                  | 125 |
| AtSAP9  | ---EVGRGEIAKANPVVKADKIQR-----                                                  | 176 |
| AtSAP10 | ---QSGRLALATQLPLIRADKLQR-----                                                  | 127 |
| AtSAP11 | SSSKDNGIVEVCPQCGAKFSSVTSLVEHVEKTHERNKKQNHGNTVDVCPRCRSGFRDPVDLVNHIERDHRGTSKA    | 279 |
| AtSAP12 | ASGNDG-----NSEKTQERNKGKQCGKVTVDVCPKCSRGRDPVDLLKHIDKDHGTSKA                     | 186 |
| AtSAP13 | EDPRVENLPLWMIREGVADVGLRGQMSQEEEDVEWEEE-----                                    | 249 |
| AtSAP14 | -----                                                                          | 191 |

**B****C**

**Fig. 2** The *Arabidopsis* SAP family. **a** Alignment of *Arabidopsis* SAP gene family encoding A20/AN1 zinc-finger proteins using clustal method. The conserved cysteine residues characteristic of the A20 zinc-finger are highlighted by medium grey colour and the conserved cysteine and histidine residues characteristic of the AN1 zinc-finger domain are marked in dark grey colour. Residues showing  $\geq 50\%$  conservation are marked in light grey colour. Gaps indicated by dashes have been introduced to maximize alignment. **b** Domain organization of the *Arabidopsis* A20/AN1 zinc-finger proteins encoded by the SAP gene family. The size of

each domain was obtained from SMART (<http://www.smart.embl-heidelberg.de/>) and InterPro database (<http://www.ebi.ac.uk/interpro/>) and is to scale with respect to the protein size. Note that AtSAP1-10 have both the A20 and AN1 zinc-finger domains while only the AN1 zinc-finger domain is present in AtSAP11-14. **c** Chromosomal localization of the *Arabidopsis* SAP gene family members. The numbers indicate the gene identity given in Table 2. Genes present in duplicated segments are connected by dashed lines

given cold ( $8 \pm 1^\circ\text{C}$ ), salt (200 mM) and dehydration (dried on tissue paper) stress treatment for 6 h and used for real time PCR. In order to gain insight into the basal expression levels of the SAP genes, the threshold cycle values (CT) of the unstressed samples (after normalization with *ACTIN*) were compared with each other. *OsSAP4*, *OsSAP6*, *OsSAP8*, *OsSAP9*, *OsSAP15* and *OsSAP16* seemed to have a high level of expression while *OsSAP12*, *OsSAP13*, *OsSAP14* and *OsSAP18* had a much lower level of expression in the unstressed state. The mRNA level of all the 18 genes (relative to the unstressed control) showed an increase in response to any one of the stress conditions used, albeit to variable extent. The fold induction in response to salt and dehydration stress was highest for *OsSAP14* (~11 folds) and *OsSAP10* (~22 folds), respectively. The rice SAP genes showed variable expression in response to cold stress. The expression of twelve of the genes (*OsSAP2*, *OsSAP3*, *OsSAP4*, *OsSAP5*, *OsSAP6*, *OsSAP7*, *OsSAP8*, *OsSAP10*, *OsSAP13*, *OsSAP15*, *OsSAP16* and *OsSAP18*) was maintained or down regulated. Thus, there were only six SAP genes (*OsSAP1*, *OsSAP9*, *OsSAP11*, *OsSAP12*, *OsSAP14* and *OsSAP17*), which showed significant increase in expression upon exposure to cold stress (Fig. 4). This included *OsSAP1*, a gene previously shown to be induced in response to multiple

abiotic stresses like cold, salt, dehydration, submergence, heavy metals, ABA and wounding (Mukhopadhyay et al. 2004).

To validate the expression pattern of the rice SAP genes under abiotic stress conditions, ~1 kb region, upstream of the open reading frame, was used to search the database for regulatory motifs using PLACE (Higo et al. 1999). Various *cis*-acting regulatory elements involved in abiotic stress responsive gene expression were identified (Table 3) indicating their association with stress inducible expression of SAP genes.

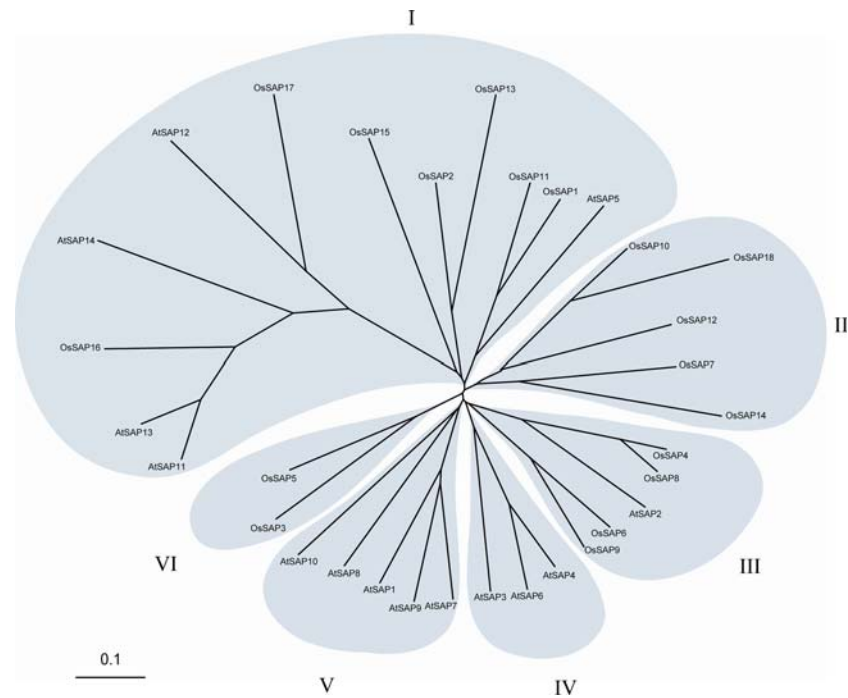
## Discussion

Proteins with zinc-fingers of A20/AN1 type play an important role in regulating the immune response in animals (Heyninck and Beyaert 2005; Dixit et al. 1990; Opipari et al. 1990; Linnen et al. 1993; Scott et al. 1998; Duan et al. 2000; Evans et al. 2004; Huang et al. 2004; Hishiya et al. 2006) and have been shown to function in stress responses in plants (Mukhopadhyay et al. 2004). Availability of the complete rice and *Arabidopsis* genome sequence (The Arabidopsis Genome Initiative 2000; The International Rice Genome Sequencing Project 2005; Vij et al. 2006) facilitated the search for

**Table 2** Distribution of *AtSAP* gene family encoding A20/AN1 zinc-finger proteins in the *Arabidopsis* genome

| S.No. | Zinc-finger       | Gene_ID        | Accession | Chr | Coordinates (5' to 3') | Nucleotide length (bp) | Predicted protein (aa) | No. of introns |
|-------|-------------------|----------------|-----------|-----|------------------------|------------------------|------------------------|----------------|
| 1     | A20-AN1           | <i>AtSAP1</i>  | AT1G12440 | 1   | 4,241,590–4,242,970    | 507                    | 169                    | 1              |
| 2     | A20-AN1           | <i>AtSAP2</i>  | AT1G51200 | 1   | 18,988,102–18,990,236  | 522                    | 174                    | 2              |
| 3     | A20-AN1           | <i>AtSAP3</i>  | AT2G27580 | 2   | 11,783,488–11,784,539  | 492                    | 164                    | 1              |
| 4     | A20-AN1           | <i>AtSAP4</i>  | AT2G36320 | 2   | 15,235,867–15,237,575  | 486                    | 162                    | 0              |
| 5     | A20-AN1           | <i>AtSAP5</i>  | AT3G12630 | 3   | 4,012,615–4,013,590    | 483                    | 161                    | 0              |
| 6     | A20-AN1           | <i>AtSAP6</i>  | AT3G52800 | 3   | 19,580,289–19,581,660  | 513                    | 171                    | 1              |
| 7     | A20-AN1           | <i>AtSAP7</i>  | AT4G12040 | 4   | 7,214,699–7,215,983    | 528                    | 176                    | 1              |
| 8     | A20-AN1           | <i>AtSAP8</i>  | AT4G14225 | 4   | 8,198,772–8,199,197    | 378                    | 126                    | 1              |
| 9     | A20-AN1           | <i>AtSAP9</i>  | AT4G22820 | 4   | 11,987,720–11,989,012  | 531                    | 177                    | 1              |
| 10    | A20-AN1           | <i>AtSAP10</i> | AT4G25380 | 4   | 12,975,946–12,976,338  | 393                    | 131                    | 0              |
| 11    | AN1-AN1-C2H2-C2H2 | <i>AtSAP11</i> | AT2G41835 | 2   | 17,464,864–17,466,632  | 840                    | 280                    | 1              |
| 12    | AN1-AN1           | <i>AtSAP12</i> | AT3G28210 | 3   | 10,521,758–10,522,769  | 561                    | 187                    | 1              |
| 13    | AN1-AN1-C2H2      | <i>AtSAP13</i> | AT3G57480 | 3   | 21,288,917–21,290,331  | 750                    | 250                    | 1              |
| 14    | AN1               | <i>AtSAP14</i> | AT5G48205 | 5   | 19,564,163–19,565,013  | 576                    | 192                    | 3              |

**Fig. 3** Phylogenetic tree of rice and *Arabidopsis* SAP gene families encoded A20/AN1 zinc-finger proteins. The unrooted tree was generated using ClustalX program by neighbour joining method and viewed by Treeview. Scale bar represents 0.1 amino acid substitutions per site. The rice and *Arabidopsis* proteins could be classified into six distinct groups (I–VI)

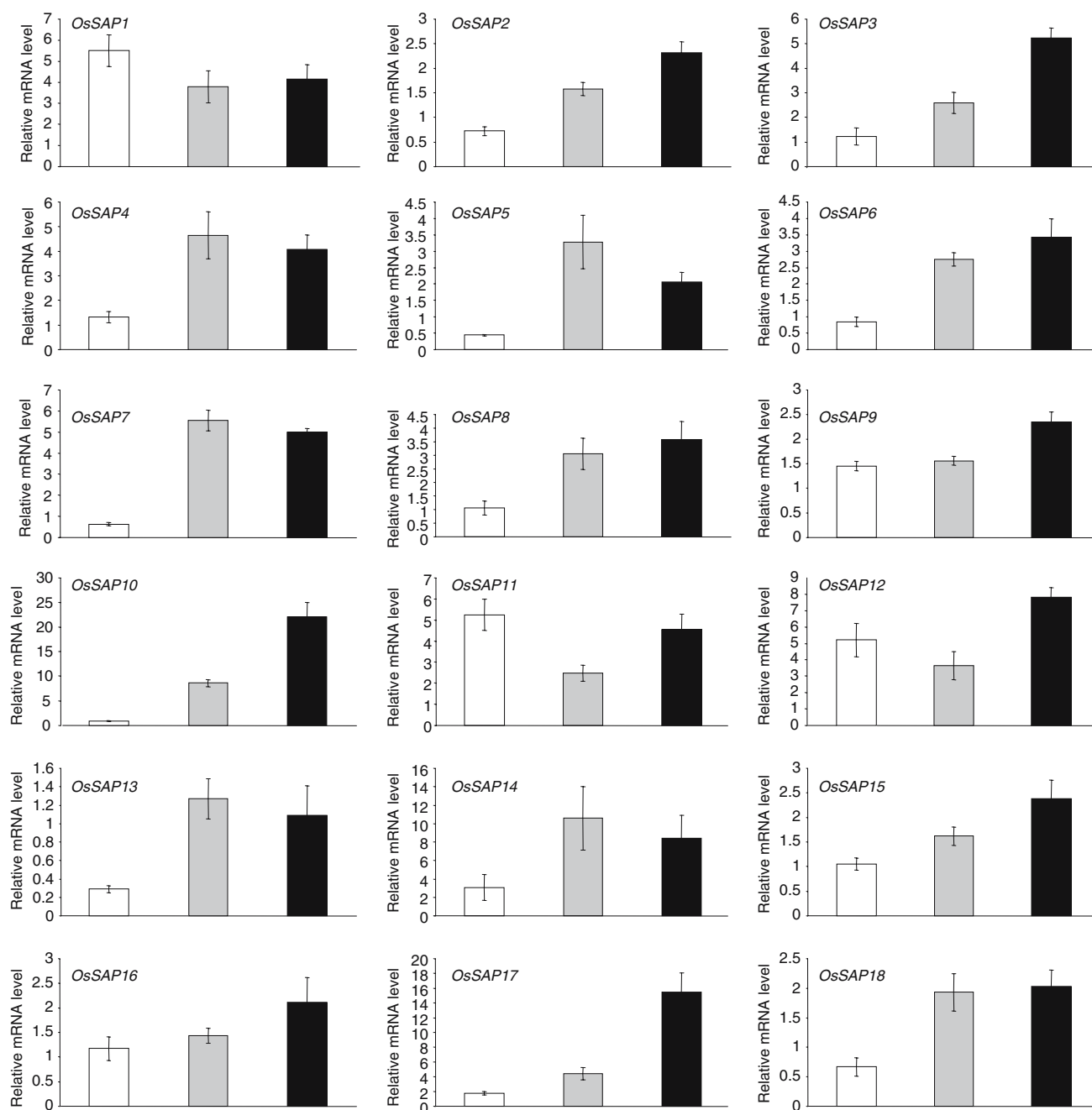


A20/AN1 zinc-finger proteins in these organisms. Due to its involvement in stress response, the first gene to be characterized for its function from *indica* rice was named as *OsiSAP1* (Mukhopadhyay et al. 2004). Eighteen and fourteen genes were identified from the rice and *Arabidopsis* genome, respectively, to code for SAP related proteins (SAP family). The majority had both the A20 zinc-finger domain (present at the N-terminus) and the AN1 zinc-finger domain (present at the C-terminus). This is consistent with the previous finding from animal systems that the A20 and AN1 zinc-finger domains are usually found associated with each other (Evans et al. 2004). In addition, there were five rice and four *Arabidopsis* proteins with only the AN1 zinc-finger domain and one rice protein with only the A20 zinc-finger domain. The presence of only the A20 zinc-finger domain is also seen in the human A20 protein (Opipari et al. 1990). On the other hand, presence of the AN1 zinc-finger domain is more prevalent across different organisms as seen by search of NCBI protein database using ‘A20 zinc-finger’ and ‘AN1 zinc-finger’ as key words. Especially, it was seen that lower organisms such as *Saccharomyces cerevisiae* and *Aspergillus fumigatus* had only the AN1 zinc-finger domain whereas A20 zinc-finger was absent indicating that the AN1 zinc-finger may be more ancient in origin compared to the A20 zinc-finger.

In contrast to the A20/AN1 zinc-finger proteins known from humans and other model animal systems, very little information is available on such proteins from plants. One such well characterized protein from

plants is the *indica* rice *OsiSAP1*. Its gene was previously shown to respond to multiple environmental stresses like cold, salt, dehydration, heavy metals, submergence, wounding as well as stress hormone abscisic acid. In addition, overexpression of *OsiSAP1* conferred tolerance to cold, salt and dehydration stress in transgenic tobacco at the seedling stage (Mukhopadhyay et al. 2004). Its paralog (*OsSAP11*) has also been used to establish protein level network during seed development and stress (Cooper et al. 2003). Besides this, an ortholog from bean is known to be inducible upon elicitor treatment and wounding stress (Sharma et al. 1992). This information available from plant as well as animal systems served as the basis to investigate genome-wide distribution of the plant SAP gene family, containing A20/AN1 zinc-finger, and their expression profile during stress. Interestingly, expression analysis of the rice *SAP* gene family showed that all the genes were inducible by one or the other abiotic stresses. In fact, all the genes showed an increase in transcript levels in response to either salt or drought stress and one third of them showed an increase in expression in response to cold stress. It was interesting to see that two rice gene pairs (*OsSAP1–OsSAP11* and *OsSAP6–OsSAP9*) present in duplicated segments were significantly similar in their expression pattern. In addition, similarity in expression pattern was also noticed for the rice genes containing only the AN1 zinc-finger domain with least expression in cold and maximum in drought. Thus, expression analysis of the rice *SAP* gene family using real-time PCR established





**Fig. 4** Real-time PCR to study the expression profile of rice *SAP* gene family members under abiotic stress conditions. The mRNA level of each gene in 7-day-old rice (*Oryza sativa* subsp. *indica* var Pusa Basmati 1) seedlings given cold (empty bars), salt (grey bars)

and dehydration (black bars) stress for 6 h was plotted relative to the value obtained for the unstressed control. The transcript level of *ACTIN* mRNA in each case served as an internal control. Standard error values are shown at the top of each bar

the association of the *SAP* gene family to abiotic stress response; however, it was difficult to place them into functional categories based on their domain organization. Although, no reports are available on functional characterization of genes coding for only the AN1 zinc-finger domain containing proteins, there are reports in humans where genes encoding proteins with only the A20 zinc-finger domain or both A20 and AN1 zinc-finger

domain have been characterized (Evans et al. 2004; Huang et al. 2004). In these cases also, it has been seen that both types of proteins (only A20 or both A20/AN1 domain containing) are implicated to have similar function (i.e. regulation of immune response) and only a detailed study involving a detailed characterization of the individual domains can throw more light on the specific role of each domain in stress response. Expression

**Table 3** Few abiotic stress related *cis* regulatory elements present in the 1 kb upstream region along with the expression pattern of rice *SAP* gene family members

| S.No | Gene ID        | ABRE | ANAEROCONSENSUS | CCAATBOX | DRE | LTRE | Cold | Salt | Dehydration |
|------|----------------|------|-----------------|----------|-----|------|------|------|-------------|
| 1    | <i>OsSAP1</i>  | +    | +               | +        | +   | +    | +++  | ++   | ++          |
| 2    | <i>OsSAP2</i>  | +    | +               | +        | —   | —    | —    | +    | ++          |
| 3    | <i>OsSAP3</i>  | —    | +               | —        | —   | —    | +    | ++   | +++         |
| 4    | <i>OsSAP4</i>  | +    | +               | +        | —   | +    | +    | ++   | ++          |
| 5    | <i>OsSAP5</i>  | +    | +               | +        | +   | +    | —    | ++   | ++          |
| 6    | <i>OsSAP6</i>  | —    | +               | +        | +   | +    | —    | ++   | ++          |
| 7    | <i>OsSAP7</i>  | +    | +               | +        | +   | +    | —    | +++  | ++          |
| 8    | <i>OsSAP8</i>  | +    | +               | +        | —   | —    | +    | ++   | ++          |
| 9    | <i>OsSAP9</i>  | +    | +               | +        | —   | —    | +    | +    | ++          |
| 10   | <i>OsSAP10</i> | +    | +               | +        | +   | +    | —    | +++  | +++         |
| 11   | <i>OsSAP11</i> | +    | +               | +        | +   | +    | +++  | ++   | ++          |
| 12   | <i>OsSAP12</i> | +    | —               | +        | +   | +    | +++  | ++   | +++         |
| 13   | <i>OsSAP13</i> | +    | +               | +        | +   | +    | —    | +    | +           |
| 14   | <i>OsSAP14</i> | +    | +               | —        | —   | +    | ++   | +++  | +++         |
| 15   | <i>OsSAP15</i> | +    | +               | +        | —   | +    | +    | +    | ++          |
| 16   | <i>OsSAP16</i> | +    | +               | +        | +   | +    | +    | +    | ++          |
| 17   | <i>OsSAP17</i> | +    | —               | +        | +   | +    | +    | ++   | +++         |
| 18   | <i>OsSAP18</i> | +    | —               | —        | +   | +    | —    | +    | ++          |

For cold, salt and dehydration stress, down regulation of expression is represented by ‘—’ (<1 folds) and upregulation is represented by ‘+’ (1–2 folds), ‘++’ (2–5 folds) and ‘+++’ (>5 folds)

of a large subset of plant genes is altered in response to stress. These include genes coding for osmoprotectants, reactive oxygen species, heat shock proteins, ion/proton transporters, water channel proteins, growth regulators, transcription factors and signaling components (Cushman and Bohnert 2000; Xiong et al. 2002; Tyagi et al. 2006). The genes involved in stress response can be broadly classified into two categories. The first is the group of genes that code for ‘regulatory’ proteins such as transcription factors and signal transduction components. The second is the group of genes that code for ‘effector’ proteins, which play a direct role in protecting the plants against stresses (Zhu 2001; Shinozaki et al. 2003). In this context, the rice *SAP* gene family encoding A20/AN1 zinc-finger proteins would probably fall in the category of regulatory genes since they code for zinc-finger domain containing proteins. Clues for possible role of *SAP* family members in rice can also be obtained from studies done with the human A20 protein. It is known that this protein can down regulate NF- $\kappa$ B (a pro-inflammatory transcription factor) through its ubiquitin ligase activity. In fact, A20-deficient mice fail to down regulate NF- $\kappa$ B activity resulting in chronic inflammation and cell death (Lee et al. 2000). A20 is able to down regulate NF- $\kappa$ B by targeting proteins involved in its signaling for degradation through the ubiquitin proteasome pathway (Wertz et al. 2004; Lee et al. 2006; Penengo et al. 2006). The possible role of the *SAP* family proteins in abiotic stress can thus be associated with this function

of the A20 zinc-finger. This zinc-finger, as in the animal systems, may be involved in down regulating the pathway associated with abiotic stress associated injuries such as cell death by ubiquitinating the key proteins and hence targeting them for degradation.

In silico analysis of the ~1 kb promoter region from rice *SAP* family members revealed several abiotic stress-responsive elements including ABA-responsive element (ABRE), coordinate regulatory element for antioxidant defense (CORE), dehydration-responsive element (DRE/CRT), low temperature responsive element (LTRE) and a heat-shock element (HSE) (Yamaguchi-Shinozaki and Shinozaki 1994; Jiang et al. 1996; Ono et al. 1996; Higo et al. 1999; Haralampidis et al. 2002; Shinozaki et al. 2003; Mukhopadhyay et al. 2004; Tsukamoto et al. 2005). However, the absence of a particular *cis* regulatory element did not always mean ‘no upregulation of expression’. This may be possibly due to the involvement of additional *cis* elements in regulating the abiotic stress response. Conversely, presence of an element did not always lead to increase in expression upon exposure to the specific abiotic stress. This may be partly due to variable kinetics of upregulation.

In conclusion, we have done genome-wide analysis of the *SAP* gene family, encoding A20/AN1 zinc-finger proteins, in rice and *Arabidopsis*. All the members present in the rice genome showed inducibility to one or the other abiotic stress. It would be important to define the relative function of the members of the gene family in the life of the rice plant.

**Acknowledgments** This work was financially supported by grants from the Department of Biotechnology (DBT), Government of India. SV acknowledges CSIR for the award of a research fellowship.

## References

- Cooper B, Clarke JD, Budworth P, Kreps J, Hutchison D, Park S, Guimil S, Dunn M, Luginbuhl P, Ellero C, Goff SA, Glazebrook J (2003) A network of rice genes associated with stress response and seed development. *Proc Natl Acad Sci USA* 100:4945–4950
- Cushman JC, Bohnert HJ (2000) Genomic approaches to plant stress tolerance. *Curr Opin Plant Biol* 3:17–124
- Dixit VM, Green S, Sarma V, Holzman LB, Wolf FW, O'Rourke K, Ward PA, Prochownik EV, Marks RM (1990) Tumor necrosis factor- $\alpha$  induction of novel gene products in human endothelial cells including a macrophage-specific chemotaxin. *J Biol Chem* 265:2973–2978
- Duan W, Sun B, Li TW, Tan BJ, Lee MK, Teo TS (2000) Cloning and characterization of AWP1, a novel protein that associates with serine/threonine kinase PRK1 in vivo. *Gene* 256:113–121
- Eddy SR (1998) Profile hidden Markov models. *Bioinformatics* 14:755–763
- Evans PC, Ovaa H, Hamon M, Kilshaw PE, Hamm S, Bauer S, Ploegh HL, Smith TS (2004) Zinc-finger protein A20, a regulator of inflammation and cell survival, has de-ubiquitinating activity. *Biochem J* 378:727–734
- Haralampidis K, Milioni D, Rigas S, Hatzopoulos P (2002) Combinatorial interaction of *cis* elements specifies the expression of the *Arabidopsis AtHsp90-1* gene. *Plant Physiol* 129:1138–1149
- Heyninck K, Beyaert R (2005) A20 inhibits NF- $\kappa$ B activation by dual ubiquitin-editing functions. *Trends Biochem Sci* 30:1–4
- Higo K, Ugawa Y, Iwamoto M, Korenaga T (1999) Plant cis-acting regulatory DNA elements (PLACE) database. *Nucleic Acids Res* 27:297–300
- Hishiya A, Iemura S, Natsume T, Takayama S, Ikeda K, Watanabe K (2006) A novel ubiquitin-binding protein ZNF216 functioning in muscle atrophy. *EMBO J* 25:554–564
- Huang J, Teng L, Li L, Liu T, Li L, Chen D, Xu LG, Zhai Z, Shu HB (2004) ZNF216 is an A20-like and IkappaB kinase gamma-interacting inhibitor of NFkappaB activation. *J Biol Chem* 279:16847–16853
- International Rice Genome Sequencing Project (2005) The map-based sequence of the rice genome. *Nature* 436:793–800
- Jiang C, Iu B, Singh J (1996) Requirement of a CCGAC *cis*-acting element for cold induction of the *BN115* gene from winter *Brassica napus*. *Plant Mol Biol* 30:679–684
- Lee EG, Boone DL, Chai S, Libby SL, Chien M, Lodolce JP, Ma A (2000) Failure to regulate TNF-induced NF-kappaB and cell death responses in A20-deficient mice. *Science* 289:2350–2354
- Lee S, Tsai YC, Mattera R, Smith WJ, Kostelansky MS, Weissman AM, Bonifacino JS, Hurley JH (2006) Structural basis for ubiquitin recognition and autoubiquitination by Rabex-5. *Nat Struct Mol Biol* 13:264–271
- Linnen JM, Bailey CP, Weeks DL (1993) Two related localized mRNAs from *Xenopus laevis* encode ubiquitin-like fusion proteins. *Gene* 128:181–188
- Mukhopadhyay A, Vij S, Tyagi AK (2004) Overexpression of a zinc-finger protein gene from rice confers tolerance to cold, dehydration, and salt stress in transgenic tobacco. *Proc Natl Acad Sci USA* 101:6309–6314
- Ono A, Izawa T, Chua NH, Shimamoto K (1996) The *rab16B* promoter of rice contains two distinct abscisic acid-responsive elements. *Plant Physiol* 112:483–491
- Opipari AW Jr, Boguski MS, Dixit VM (1990) The A20 cDNA induced by tumor necrosis factor alpha encodes a novel type of zinc finger protein. *J Biol Chem* 265:14705–14708
- Penengo L, Mapelli M, Murachelli AG, Confalonieri S, Magri L, Musacchio A, Di Fiore PP, Polo S, Schneider TR (2006) Crystal structure of the ubiquitin binding domains of Rabex-5 reveals two modes of interactions with ubiquitin. *Cell* 124:1183–1195
- Scott DA, Greinwald JH Jr, Marietta JR, Drury S, Swiderski RE, Vinas A, DeAngelis MM, Carmi R, Ramesh A, Kraft ML, Elbedour K, Skworak AB, Friedman RA, Srikumari Srisailapathy CR, Verhoeven K, Van Camp G, Lovett M, Deininger PJ, Batzer MA, Morton CC, Keats BJ, Smith RJH, Sheffield VC (1998) Identification and mutation analysis of a cochlear-expressed, zinc finger protein gene at the DFNB7/11 and *dn* hearing-loss-loci on human chromosome 9q and mouse chromosome 19. *Gene* 215:461–469
- Sharma YK, Hinojos CM, Mehdy MC (1992) cDNA cloning, structure and expression of a novel pathogenesis-related protein in bean. *Mol Plant Microbe Interact* 5:89–95
- Shinozaki K, Yamaguchi-Shinozaki K, Seki M (2003) Regulatory network of gene expression in the drought and cold stress responses. *Curr Opin Plant Biol* 6:410–417
- The Arabidopsis Genome Initiative (2000) Analysis of the genome sequence of the flowering plant *Arabidopsis thaliana*. *Nature* 408:796–815
- Thompson JD, Gibson TJ, Plewniak F, Jeanmougin F, Higgins DG (1997) The CLUSTAL\_X windows interface: flexible strategies for multiple sequence alignment aided by quality analysis tools. *Nucleic Acids Res* 25:4876–4882
- Tsukamoto S, Morita S, Hirano E, Yokoi H, Masumura T, Tanaka K (2005) A novel *cis*-element that is responsive to oxidative stress regulates three antioxidant defense genes in rice. *Plant Physiol* 137:317–327
- Tyagi AK, Vij S, Saini N (2006) Functional genomics of stress tolerance. In: Rao KVM, Raghavendra AS, Reddy KJ (eds) *Physiology and molecular biology of stress tolerance*. Springer, Dordrecht, pp 301–334
- Vij S, Gupta V, Kumar D, Vydianathan R, Raghuvanshi S, Khurana P, Khurana JP, Tyagi AK (2006) Decoding the rice genome. *BioEssays* 28:421–432
- Wertz IE, O'Rourke KM, Zhou H, Eby M, Aravind L, Seshagiri S, Wu P, Wiesmann C, Baker R, Boone DL, Ma A, Koonin EV, Dixit VM (2004) De-ubiquitination and ubiquitin ligase domains of A20 downregulate NF- $\kappa$ B signaling. *Nature* 430:694–699
- Xiong L, Schumaker KS, Zhu JK (2002) Cell signaling during cold, drought, and salt stress. *Plant Cell* 14:65–183
- Yamaguchi-Shinozaki K, Shinozaki K (1994) A novel *cis*-acting element in an *Arabidopsis* gene is involved in responsiveness to drought, low-temperature, or high-salt stress. *Plant Cell* 6:251–26
- Zhu JK (2001) Cell signaling under salt, water and cold stresses. *Curr Opin Plant Biol* 4:401–406